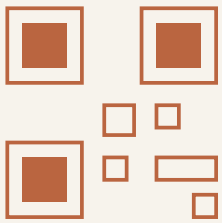
AROGYASETU · PATIENT CASE-TAKING SOFTWARE FOR INDIAN OPDS

# The doctor should listen. The software should write.

*From a seven-minute verbal complaint in any Indian language — to a structured, coded, shareable medical record — in a single OPD visit, driven by one QR code. Bilingual by design: Ayurveda & Allopathy in one record.*



Interactive deck · ←/→ to navigate · P to export PDF · live demo on localhost:3000

01 · THE PROBLEM, IN THE REAL WORLD

# The consultation is the most valuable two minutes in Indian healthcare. Half of it is spent writing.

**~2 min**

Average doctor–patient consultation time in India — among the shortest of 67 countries studied

**1 : 834**

Government-reported doctor : population ratio — a doctor for every 834 people, ~80% of whom never reach a hospital record system

**5×**

The same history is retold at five desks — reception, nurse, doctor, lab, pharmacy — because nothing carries forward

**Paper**

Prescriptions, lab reports and discharge summaries live in plastic bags; a returning patient is, digitally, a stranger

**Sources:** Irving et al., *BMJ Open* 2017 (international consultation-time study, 67 countries, 2012–13 NCDRHS India data) · PIB / Ministry of Health & Family Welfare 2024 (doctor availability) · field observation at Indian OPDs.

02 · THE BARRIER NOBODY BUILDS FOR

# Healthcare in India speaks a language many patients don't read.

A tribal patient in Chamarajanagar can explain their illness fluently — but cannot fill an English intake form. Studies at Indian district hospitals document patients leaving OPDs without understanding their diagnosis or medicines. Forms exclude; voice includes.

**22**

Scheduled languages, 100+ major mother tongues — India has no single "national" clinical language

**~40%**

Of Indian adults cannot read a standard prescription label comfortably

**22**

Languages our speech pipeline understands (Sarvam Saaras v3) — the patient speaks; the record is written for the doctor

**Sources:** Census of India (scheduled languages) · UNESCO Institute for Statistics, India literacy profile · Karanth & Vaid, language barriers in Karnataka district-hospital care, 2022.

03 · WHY NOW

# The rails are already laid — India built the world's largest digital health network.

**73.9 crore**
ABHA digital health accounts created under Ayushman Bharat Digital Mission

**49 crore**
Health records already linked to ABHA — but most OPD software still can't feed them structured data

**50 crore+**
Teleconsultations on eSanjeevani — proof that Indians will use digital health at population scale

**$43.4 Bn**
AYUSH & herbal market size (2024) — 3,000+ government Ayush hospitals with no modern case-taking software

The infrastructure (identity, consent, records) exists. What's missing is the **point-of-care layer**: software that captures the consultation itself. That is ArogyaSetu.

**Sources:** PIB, Ministry of Health & Family Welfare — ABDM milestones (Feb 2025); eSanjeevani 5-crore+ milestone updates (2025–26); IMARC India AYUSH & Herbal Market Report 2024.

04 · THE SOLUTION

# ArogyaSetu — one QR code, one conversation, one complete medical record.

### One QR, whole hospital

Booking → check-in → vitals → consult → pharmacy → billing → discharge. Every desk scans the same code; the history is never retold.

### The patient talks, the app writes

Voice-first intake and ambient scribe convert speech in 22 languages into SOCRATES-structured notes and SOAP drafts the doctor approves.

### Ayurveda is computable

Prakriti, Ashtavidha and Dashavidha assessments, plus 3-way dual coding: NAMASTE ↔ ICD-11 TM2 ↔ ICD-11 biomedicine.

### ABDM-ready by design

ABHA linking, FHIR R4 document bundles and care-context references for every encounter — exportable to the national grid.

8 role-based portals · patient pre-consult intake · encrypted medical Drive · family case files · referral dispatch · drug-interaction safety · FHIR export

05 · THE HERO FLOW

# Ten steps, five desks, one code. *Nothing is re-entered, ever.*

| | | | | |
|---|---|---|---|---|
| 01<br>**PATIENT**<br>Books appointment in 6 guided steps | 02<br>**PATIENT**<br>Gets QR slip `AIIA–CK–…` + queue token | 03<br>**PATIENT**<br>Pre-consult intake by voice, consent-gated | 04<br>**RECEPTION**<br>Scans QR → checked in (the only check-in path) | 05<br>**NURSE**<br>Scans QR → vitals recorded |
| 06<br>**DOCTOR**<br>Queue auto-promotes red-flag / emergency cases | 07<br>**DOCTOR**<br>Consultation screen — intake already filled | 08<br>**PHARMACY**<br>Scans QR → dispenses, stock + bill auto-raised | 09<br>**BILLING**<br>One click pay — consult + meds + labs | 10<br>**DOCTOR**<br>Discharge → FHIR bundle + follow-up booked |

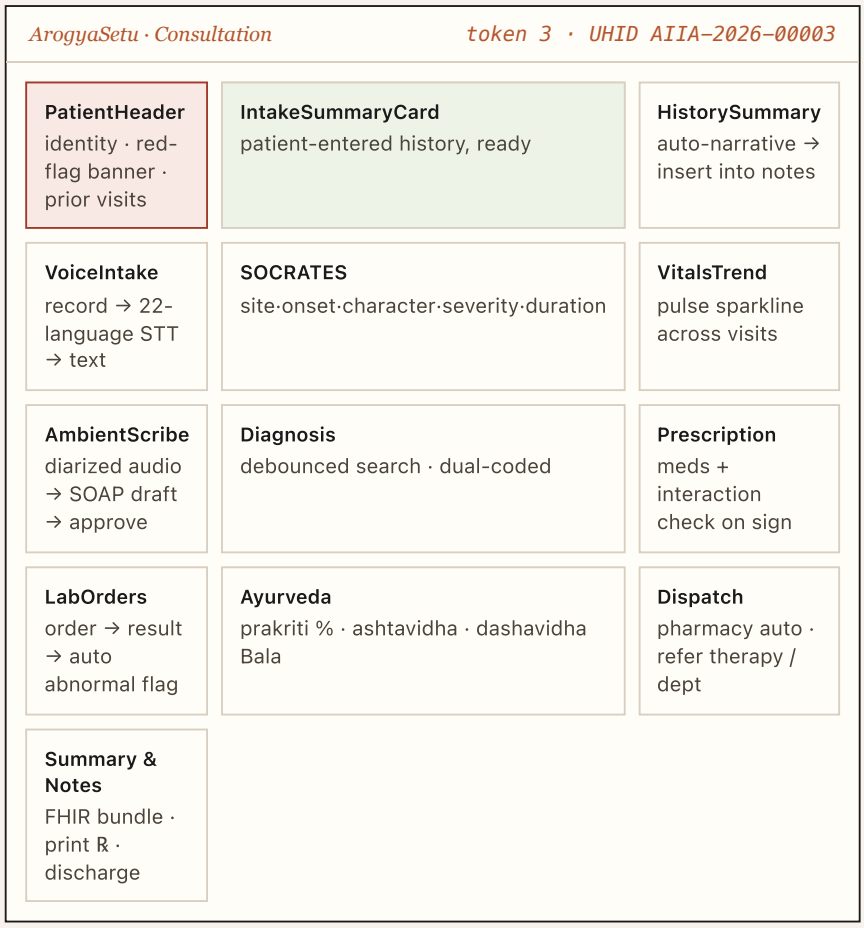
Row-locked token allocation (no double-booking) · forged or guessed codes return **404** · walk-in & emergency patients enter at the front of the queue with token E

06 · INSIDE THE CONSULTATION

# Thirteen cards. Everything the

# doctor needs, nothing they had to type.

- **Intake arrives pre-filled** — complaints, SOCRATES history, allergies, medicines, past history, positive ROS — collected from the patient before they left home
- **Red-flag safety** — "chest pain / breathlessness / bleeding" in the complaint auto-flags the case and promotes it in the queue
- **One pick, three codes** — choosing *Amlapitta* records NAM-AYU-0001 + TM26.SA00 + K29.7 simultaneously
- **Sign with confidence** — prescriptions are interaction-checked against a



severity pair-list before signing

07 · THE AI THAT DOES THE TYPING

# Voice in, structure out.

**Patient speaks**
"Do din se seene mein jalan hai, khane ke baad badh jaati hai..."

→

**Sarvam Saaras v3 STT**
22 Indic languages, diarized two-speaker consultation audio

→

**SOCRATES extractor**
Site · Onset · Character · Radiation · Associations · Timing · Severity · Duration

→

**SOAP draft**
LLM drafts Subjective / Objective / Assessment / Plan

→

**Doctor approves**
One click — the doctor remains the signer of record

**Pre-consult intake (patient, at home):** consent-gated adaptive questions — the branching engine scores complaint keywords and asks only what matters (cardiac ≠ dermatology).

**In the room (doctor):** ambient scribe records the conversation, separates the two speakers, drafts the note; vitals trend and prior visits are one glance away.

**Paper → digital:** patients photograph old lab reports / prescriptions; OCR + extraction recovers values, flags abnormals against reference ranges, and the doctor confirms.

08 · THE PRINCIPLE JUDGES REMEMBER

# Honest AI — the system never invents a medical fact.

**1 · Real — Sarvam AI**
Primary engine per capability (STT, translate, chat, TTS, OCR). Key held encrypted, admin-configurable at runtime.

→

**2 · Fallback — local**
openai-whisper and Ollama run on the same machine. Works offline; no cloud dependency for core capture.

→

**3 · Labelled mock**
Only if both fail. Response is flagged `mocked:true` with the real error attached. Never fabricates text.

- **Tested honesty:** the suite includes an explicit assertion that when no engine can transcribe, the API returns `provider:"none"` + error — not a made-up transcript
- **Admin transparency:** a live real-vs-mocked table on the admin portal shows exactly which capabilities are live
- **Mocked by design (and labelled):** ABDM sandbox bridge (OTP 123456), payment gateway
- **Human stays in charge:** AI drafts; the doctor reads, edits, and approves every note and prescription

09 · MAKING 3,000 YEARS COMPUTABLE

# One diagnosis, three codes — Ayurveda and biomedicine in the same record.

### Doctor picks the Ayurvedic term

Debounced search over 42 curated NAMASTE terms with synonyms (Amlapitta, Jwara, Shwasa, Kasa, Prameha, Sandhivata…)

→

### 3 rows written at once

NAMASTE `NAM-AYU-0001` · ICD-11 TM2 `TM26.SA00` · ICD-11 biomedicine `K29.7` (reflux oesophagitis / gastritis)

→

### FHIR R4 bundle

Dual-coded Conditions + prakriti Observations in a document Bundle — legible to any ABDM-participating hospital

### Prakriti & Vikriti

18-trait questionnaire scores Vata / Pitta / Kapha percentages; current symptoms identify the aggravated dosha.

### Ashtavidha & Dashavidha

The 8 classical examinations (pulse, tongue, urine…) and Charaka's ten-fold assessment with a computed Bala (strength) score.

### Why it matters

India has 3,000+ Ayush hospitals and 28,000+ dispensaries whose records are invisible to the biomedical system. Dual coding ends that silo.

10 · PRIVACY IS THE PRODUCT

# Encrypted, consented, audited — and the audio deletes itself.

### AES · Fernet

Aadhaar numbers encrypted at rest, masked XXXX–XXXX–1234 in every API response. Medical Drive blobs stored encrypted, never publicly served.

### Rotating sessions

Short-lived JWT + httpOnly refresh cookie that rotates on every use; patients can see and revoke all active sessions.

### Granular consent

Consultation recording requires explicit per-item consent, is revocable, and raw audio auto-purges after the retention window (default 30 days).

### Audit & ownership

Every sensitive action logged; patients can read only their own records (403-tested), staff act only within their role.

Production mode refuses to boot with a development secret or missing encryption key · drug-interaction pair list guards prescriptions · access to another patient's Drive documents requires an explicit, expiring, reason-tagged grant

11 · UNDER THE HOOD

# Two clean services, standard protocols, nothing exotic to deploy.

*System architecture* *localhost:3000 / :8000*

**Next.js 14 · React 18 · Tailwind · zustand**
34 routes, 8 role portals · jsQR camera scanning · QR generation · 6 UI languages · dark mode · 401→silent refresh

**FastAPI · SQLAlchemy 2 async · Pydantic v2**
128 endpoints · 11 routers · SQLite (Postgres-ready) · lifespan auto-migration & seed · Swagger at /docs

**Domain services**
SOCRATES NLP · prakriti / dashavidha scoring · terminology 3-way mapping · adaptive question engine · OCR & lab extraction · FHIR R4 builder + validator · history summariser

**AI chain**
Sarvam AI (STT·translate·chat·TTS·OCR) → Whisper / Ollama local → labelled mock

**Data**
39 tables · encrypted blob store `media/drive/{uuid}.bin` · audit log · retention engine

— **128 API endpoints**, every one role-guarded and covered by an interactive Swagger UI — judges can poke the API themselves

— **33 automated tests** exercise the full 8-role journey, cross-desk data flow, security guards and AI honesty

— **Docker Compose** included; SQLite today, Postgres-ready ORM for scale

— **Open standards only** — FHIR R4, JWT, OAuth-style refresh, ABHA/ABDM patterns. No proprietary lock-in.

12 · BY THE NUMBERS

# Built end-to-end, not a mock-up.

| | | | | | |
|---|---|---|---|---|---|
| **39** database tables | **128** API endpoints | **34** frontend routes | **33** automated tests, all passing | **8** role-based portals | **22** speech-to-text languages |

**42** curated terminology entries → **126** coded rows across three systems

**13** interactive cards on the consultation screen · **6** UI languages

**2** real AI engines + honest fallback chain · **1** QR code for the entire visit

13 · REAL-WORLD IMPACT

# Where this goes next — the point of the whole exercise.

### Primary Health Centres

A typical PHC doctor sees 100+ patients a day with no digital record. ArogyaSetu turns each visit into structured, coded, FHIR-exportable data — feeding ABHA, not plastic bags.

### Ayushman Arogya Mandirs

India's 1.5+ lakh Ayush health & wellness centres deliver integrative care that no EHR represents today. Dual-coded records make AYUSH outcomes measurable nationally.

### Continuity of care

A migrant worker treated in Bengaluru walks into a PHC in Bihar; the case file, allergies and medicines follow the ABHA account, not the paper.

### Public-health signal

When every consultation is coded (ICD-11 / NAMASTE), disease surveillance moves from annual surveys to near-real-time — this is the A, of ABDM's own "Arogya Setu" spirit.

Alignment: UN SDG-3 (good health & well-being) · Ayushman Bharat Digital Mission · National AYUSH Mission · Digital India. Deployable as a single container per facility; Sarvam-first with fully offline Whisper/Ollama mode for low-connectivity districts.

14 · LIVE DEMO

# Everything shown is running on this machine, right now.

- **Patient portal** — `localhost:3000/login` · patient@demo.in
- **Staff portal** — `localhost:3000/staff/login` · doctor-allo@demo.in (all 8 roles one-click)
- **API playground** — `localhost:8000/docs` · 128 endpoints, live
- **Source code** — `github.com/venkaraddi-crypto/qriuos` · 33 tests: `pytest tests/ -q`
- **Password for every demo account** — `password123`

## 7-minute demo path

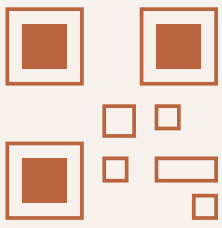
1. Patient books → QR slip (30s)
2. Voice intake, consent-gated (90s)
3. Reception scans QR (30s)
4. Nurse vitals (20s)
5. Doctor: queue → consult → voice → SOCRATES → Amlapitta → 3 codes → prescribe → FHIR (3min)
6. Pharmacy dispense → Billing pay (45s)
7. Patient sees records + case file (30s)

AROGYASETU · आरोग्यसेतु — "THE BRIDGE TO HEALTH"

# The record should be the bridge, not the barrier.

*One QR code connects every desk. One conversation fills the record.*
*Two systems of medicine share one language of codes. And the doctor — finally — just listens.*

---



Thank you · github.com/venkaraddi-crypto/qriuos · live demo available now